



COP4710

Database Systems

Introduction

Tallahassee, Florida

Welcome to COP4710



- **Course Website:**
 - <http://www.cs.fsu.edu/~zhao/cop4710/main.html>
 - Syllabus, policies, schedules, slides, projects, resource...
- **Canvas**
 - Announcements, homework, grades, submissions
- **Please read the class syllabus, policies, and lecture schedule carefully; ask now if you have questions**
 - **No recitations for this class**
 - Please make good use of office hours of TAs/Instructor's
 - **FSU first day attendance policy**

Teaching Staff



- **Instructor: Peixiang Zhao** (zhao@cs.fsu.edu)
 - **Research interest**
 - Generally, **data science** including **database systems** and **data mining**
 - Specifically, **graph** data, information network analysis, large-scale data-intensive computation and analytics
 - **Brief history**
 - Illinois (Ph.D. from UIUC, 2012)
 - FSU (Professor, at FSU since 2012)
- **TA:**
 - Exceptional graduate students here at FSU
 - ???

You Tell Me --



- **Why Are You Taking this Course?**
 - CS or Non-CS? Any experiences in RDBs, NoSQL?
- **Are you interested more in being**
 - An **IT guru** at Goldman-Sachs or Boeing?
 - A **system developer** at Oracle or Google?
 - A **data scientist** at Meta or Uber?
 - A **DB pro or researcher** in Microsoft research?
 - A **professor** exploring one of the most exciting, and fastest growing areas in CS?

Examples



Off Campus? Log in to access full text and more content.

LIBRARIES database system [Advanced Search](#)

☐ Keep search refinements ☒ New search

Search Results: Your search for **database system** returned **2,993,243** results

Refine your search

- ☐ Items with full text online
- ☐ Limit to articles from scholarly publications, including peer-review
- ☐ Exclude Newspaper Articles
- ☐ Add results beyond your library's collection

Content Type

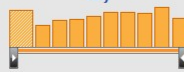
- ☒ Any
- ☐ Journal Article (1,603,414)
- ☐ Newspaper Article (1,105,753)
- ☐ Dissertation/Thesis (124,873)
- ☐ Conference Proceeding (56,148)
- ☐ Trade Publication Article (54,341)
- ☐ Book / eBook (45,131)
- [more...](#)

Subject Terms

- ☒ Any
- ☐ studies (167,538)
- ☐ engineering, electrical & electronic (71,620)
- ☐ analysis (70,062)
- ☐ united states (67,609)
- ☐ research (64,844)
- ☐ computer science (63,775)
- [more...](#)

Publication Date

Any



2012 Search For Journal Articles, Books, and More | Powered by **Summon**

Database system: design and development

by Kashyap, M. M.
1993, ISBN 8120712633, x, 382 p.
Database design, Automation, Database management, Computer networks, Microcomputers

Mobile database systems

by Kumar, Vijay
2006, ISBN 9780471467922, xxiv, 294 p.
...Contents Acknowledgments xxi Preface xxiii 1 Mobile **Database System** 1.1 introduction 1.1.1 Fully Connected information Space 1.2 Types of Mobility 1.3 Summary... Mobile communication systems, Mobile computing

PathCase: pathways database system

by Elliott, Brendan; Kirac, Mustafa; Cakmak, Ali; Yavas, Gokhan; Mayes, Stephen; Cheng, En
...(more)
Bioinformatics, ISSN 1367-4803, 11/2008, Volume 24, Issue 21, pp. 2526 - 2533
...BIOINFORMATICS ORIGINAL PAPER Vol. 24 no. 21 2008, pages 2526-2533
doi:10.1093/bioinformatics/btn459 **Databases** and ontologies PathCase: pathways **database system**...

Real-time database systems: architecture and techniques

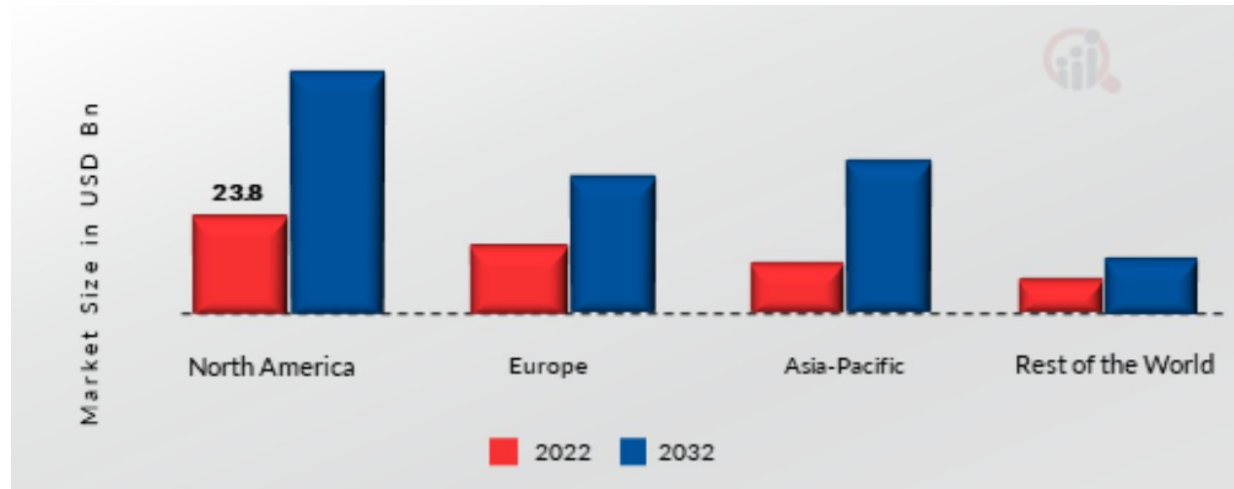
by Lam, Kam-Yiu; Kuo, Tei-Wei; NetLibrary, Inc
2001, The Kluwer international series in engineering and computer science, ISBN 9780306469886, Volume SECS 593
Database design, Real-time data processing

Nontraditional database systems

by Kambayashi, Y and Jôhō Shori Gakkai (Japan)
2002, Advanced information processing technology, ISBN 9780415302067, Volume 5, xvii, 251 p.
"The book examines non-traditional types of applications, data types, systems and environments"

[Saved Items](#)

In Industry



In Industry



423 systems in ranking, January 2025

Rank			DBMS	Database Model	Score		
Jan 2025	Dec 2024	Jan 2024			Jan 2025	Dec 2024	Jan 2024
1.	1.	1.	Oracle +	Relational, Multi-model i	1258.76	-5.03	+11.27
2.	2.	2.	MySQL +	Relational, Multi-model i	998.15	-5.61	-125.31
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model i	798.55	-7.14	-78.05
4.	4.	4.	PostgreSQL +	Relational, Multi-model i	663.41	-2.97	+14.45
5.	5.	5.	MongoDB +	Document, Multi-model i	402.50	+2.12	-14.98
6.	↑ 7.	↑ 9.	Snowflake +	Relational	153.90	+6.54	+27.98
7.	↓ 6.	↓ 6.	Redis +	Key-value, Multi-model i	153.36	+3.08	-6.03
8.	8.	↓ 7.	Elasticsearch	Multi-model i	134.92	+2.60	-1.15
9.	9.	↓ 8.	IBM Db2	Relational, Multi-model i	122.97	+0.19	-9.43
10.	10.	↑ 11.	SQLite	Relational	106.69	+4.97	-8.51

In Science – Turing Awardees



CHARLES BACHMAN, 1973



EDGAR CODD, 1981



JAMES GRAY, 1998



MICHAEL STONEBRAKER,
2014



JEFFREY ULLMAN,
2020

COP4710 Goal



1. How to **use** a database system?

- Conceptual data modeling, the relational and other data models, database schema design, relational algebra, and the SQL query language
-

2. How to **design and implement** a database system?

- Indexing, transaction processing, and crash recovery
-

Prerequisite



- **Must** have data structure and algorithm background
 - **COP3330**: Object-oriented Programming and **MAD2104**: Discrete Mathematics
 - or equivalent
- **Good programming skill**
 - Project will require lots of programming
 - Need C++, Java, PHP, Python, ... to do a good job at talking with DB
 - **You or your project group picks the topic and language**

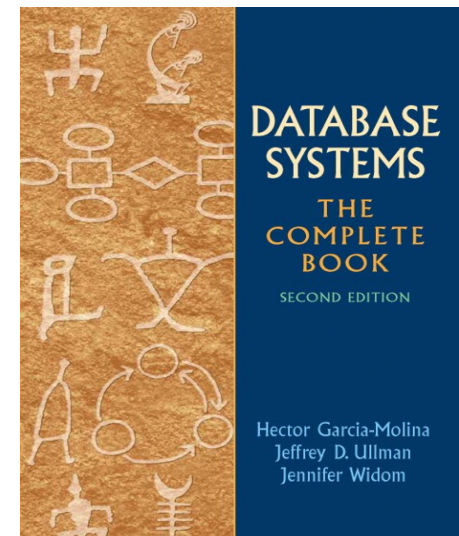
Textbook



- **Database Systems: The Complete Book. 2nd edition**
 - <http://infolab.stanford.edu/~ullman/dscb.html>

- **References**

- Database Management Systems
- Database system concepts
- Fundamentals of Database Systems



Course Format



- **Two 75-min lectures/week**
 - Lecture slides are used to complement the lectures, not to substitute the textbook
- **Four assignments planed (30%)**
 - Individual work
 - No late homework will be accepted
- **A programming project (30%)**
 - Teamwork (at most 3 students)
 - Multi-stage tasks involving a lot of programming
- **One midterm (10%) and one final (30%)**
 - Check dates and make sure no conflict!

Project



- An **RDB-driven Web-based** information system
 - Select a real-world application that needs **databases** as backend systems
 - Design and build it from start to finish
 - Your choice of topic: useful, realistic, **database-driven, Web-based**
- **Requirement**
 - Teamwork (at most three students)
 - all members receive the same scores, and if one drops out, the others pick up the work
 - Will be done in stages
 - you will submit some deliverables at the end of each stage
 - Will **show a demo** and **submit a report** near the

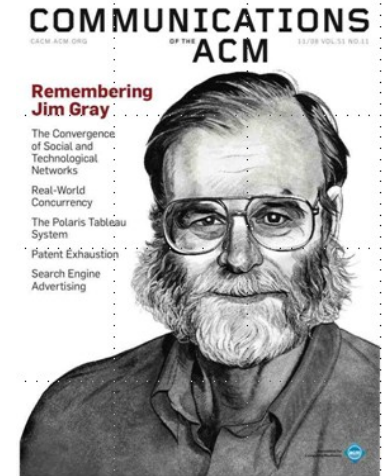
Data Management Evolution



Jim Gray: *Evolution of Data Management.*

IEEE Computer 29(10): 38-46 (1996):

- Manual processing: 4000BC --
- Mechanical punched-cards: 1900-1955
- Stored-program computer -- sequential record processing: 1955-1970
- Online navigational network DBs: 1965-1980
 - many applications still run today!
- **Relational DB: 1970-**
- **Post-relational and the Internet: 1995-**



Big Data Landscape 2021



Big Data Landscape



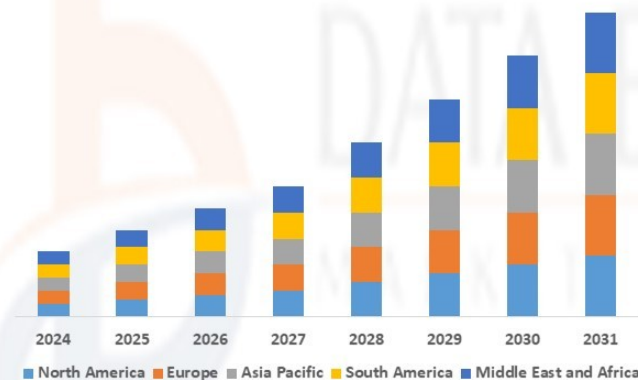
Graph Databases



The graph database ecosystem 2019



Global Graph Database Market is Expected to Account for USD 8.72 Billion by 2031

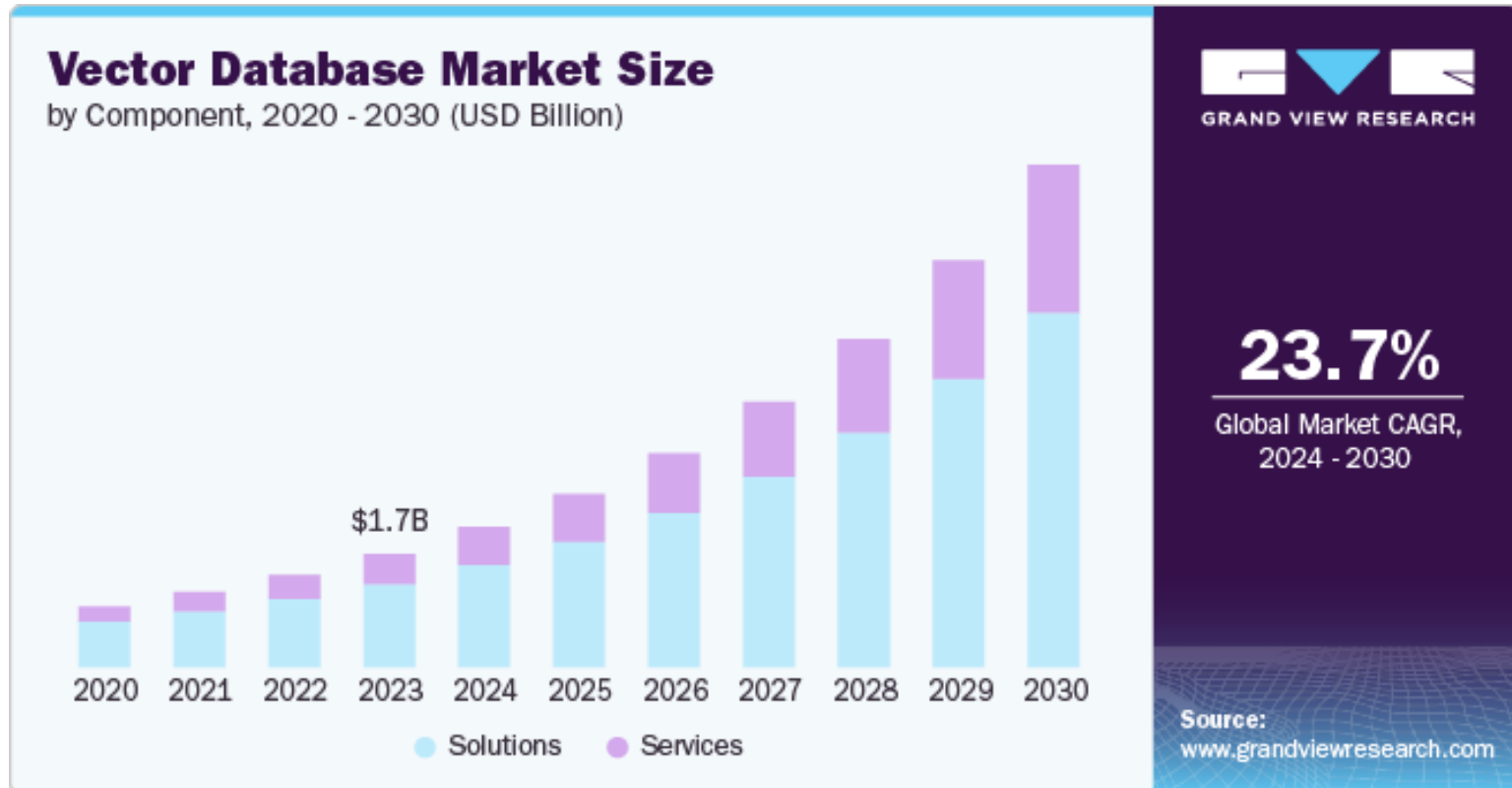


Global Graph Database Market, By Regions, 2024 to 2031



DATA BRIDGE MARKET RESEARCH

Vector Databases



Database Management System (DBMS)



System for providing **EFFICIENT, CONVENIENT**, and **SAFE MULTI-USER storage** of and **access** to **MASSIVE** amounts of **PERSISTENT data**

Example: Banking System



- **Data**
 - Information on accounts, customers, balances, current interest rates, transaction histories, etc.
- **MASSIVE**
 - many gigabytes at a minimum for big banks, more if keep history of all transactions, even more if keep images of checks -> Far too big for memory
- **PERSISTENT**
 - data outlives programs that operate on it

Example: Banking System



- **SAFE:**
 - from system failures
 - from malicious users
- **CONVENIENT:**
 - simple commands to - debit account, get balance, write statement, transfer funds, etc.
 - also unpredicted queries should be easy
- **EFFICIENT:**
 - don't search all files in order to get the balance of one account, get all accounts with low balances, get large transactions, etc.
 - massive data! -> DBMS's carefully tuned for performance

Multi-user Access



- Many people/programs accessing same database, or even same data, simultaneously -> Need careful controls
 - Alex @ ATM1 (New York): withdraw \$100 from account #007
 - get balance from database;**
 - if balance $\geq$ 100 then balance := balance - 100;
 - dispense cash;
 - put new balance into database;
 - Bob @ ATM2 (Tallahassee): withdraw \$50 from account #007
 - get balance from database;**
 - if balance $\geq$ 50 then balance := balance - 50;
 - dispense cash;
 - put new balance into database;
 - **Initial balance = 200. Final balance = ??**

Why File Systems Won't Work



- **Storing data: file system is limited**
 - size limit by disk or address space
 - when system crashes we may lose data
 - Password/file-based authorization insufficient
- **Query/update:**
 - need to write a new C++/Java program for every new query
 - need to worry about performance
- **Concurrency: limited protection**
 - need to worry about interfering with other users
 - need to offer different views to different users (e.g. registrar, students, professors)
- **Schema change:**
 - entails changing file formats
 - need to rewrite virtually all applications

That's why the notion of DBMS was motivated!

DBMS Architecture



User/Web Forms/Applications/DBA

query

transaction

DDL commands

Query Parser

Transaction Manager

DDL Processor

Query Rewriter

Query Optimizer

Concurrency
Control

**Logging &
Recovery**

Query Executor

Records

Indexes

Buffer Manager

Storage Manager

Buffer:
data, indexes, log, etc

Lock Tables

Main Memory

Storage

data, metadata, indexes, log, etc

Data Structuring: Model, Schema, Data



- **Data model**
 - conceptual structuring of data stored in database
 - ex: data is set of records, each with student-ID, name, address, courses, photo
 - ex: data is graph where nodes represent cities, edges represent airline routes
- **Schema versus data**
 - **schema**: describes how data is to be structured, defined at set-up time, rarely changes (also called "**metadata**")
 - Otherwise, you have random bits with no meaning
 - **data** is actual "**instance**" of database, changes rapidly
 - vs. types and variables in programming languages

Data Models



Relational

← Most DBMSs

Key/Value

← Simple Apps / Caching

Graph

Document/Json/XML/Object

← NoSQL

Wide-column/Column-family

Array(Vector, Matrix, Tensor)

← ML / Science

Hierarchical

Network

← Obsolete / Legacy

Semantic

Entity-Relationship

← Conceptual Modelling

Schema vs. Data



- **Schema:** name, name of each field, the type of each field
 - Students (Sid:string, Name:string, Age: integer, GPA: real)
 - A template for describing a student
- **Data:** an example instance of the relation

Sid	Name	Age	GPA
0001	Alex	19	3.55
0002	Bob	22	3.10
0003	Chris	20	3.80
0004	David	20	3.95
0005	Eugene	21	3.30

Data Structuring: Model, Schema, Data



- **Data definition language (DDL)**
 - commands for setting up schema of database
- **Data Manipulation Language (DML)**
 - Commands to manipulate data in database:
 - RETRIEVE, INSERT, DELETE, MODIFY
 - Also called "query language"

People



- **DBMS user: queries/modifies data**
- **DBMS application designer**
 - set up schema, loads data, ...
- **DBMS administrator**
 - user management, performance tuning, ...
- **DBMS implementer: builds systems**

How to Get the Most out of COP4710?



- **Read and think before class**
 - welcome to ask questions before/during/after class!
- **Study and discuss with your peers**
 - discuss readings to enhance understanding
 - **discuss assignments but write your own solution!**
- **Use lectures to guide your study**
 - use it as a roadmap for what's important
 - **lectures are starting points– they do not cover everything you should read**
- **Participate actively in your project**

Questions

